

Car Accident Trends in the U.S. Visualized

Group 3: Kira Damo, Massiel Payero Garcia, Katrina Grigorovica, Zofia Landowska,
Alexander Williams

DSC 465

Introduction:

According to the Centers for Disease Control, over 2.8 million people visited the emergency room in 2023 due to a car accident-related injury¹. Of these injuries, over 44,000 people did not survive, with about 120 people dying daily¹. Car accidents are one of the top causes of death in the United States, costing \$457 billion annually. Despite the evolution of vehicle safety, new technology, and transportation improvements, motor vehicle accidents continue to be a major problem. Our project aims to use visualizations to further understand and convey the circumstances and primary influences behind car accidents, while providing actionable insights to mitigate risk and reduce accident rates in the United States. Our analysis reveals that accident risk is not random, following predictable patterns in time, location, and conditions, suggesting clear opportunities for intervention.

The data analyzed in this project is from *US Accidents (2016 - 2023)* sourced from Kaggle². The dataset consists of 7.7 million car accidents from across the United States, describing 46 different features for each accident. For our project, a randomly sampled subset, also found on Kaggle, of 500,000 accidents was used. The data consisted of 46 various features related to time, location, weather conditions, environmental factors, severity, and other variables that describe each reported accident. Our exploration will focus on generating insightful visualizations relating to geographic trends, temporal patterns, and severity levels in car accidents.

Exploratory Analysis:

To identify meaningful patterns in the U.S. Car Accidents dataset, we conducted exploratory analyses across temporal, geographic, and environmental dimensions, examining how frequency and severity varied across each. The visualizations captured key findings and insights that helped us shape our investigative directions and revealed the most impactful and interpretable patterns in the data.

Initially, we explored the temporal features in the dataset. Histograms were used to visualize hourly and daily trends in the data, seen Figure 1 of Appendix B. The histogram of accidents by hour, seen on the left of Figure 1, shows a bimodal distribution peaking during morning (7AM-9AM) and evening (4PM-6PM) rush hours. The histogram of accidents by day

of the week, seen on the right of Figure 1, shows accidents occur less over the weekend than during weekdays. Considered together, these plots suggest that commuter traffic volume, by time and weekday, is an important driver of accident frequency. This finding directed us toward temporal patterns as a focus of final visualizations.

Next, we explored the geographic and locational features of the dataset. Choropleths and bar charts were used to visualize what states recorded the most car accidents, as seen in Figure 2 of Appendix B. The choropleth of accident count by U.S. state and the bar chart of the top 10 U.S. states with the most accidents in Figure 2 show what states recorded the most accidents in our dataset. Both plots reveal that accident counts appear to closely follow population, as the states with the largest population have the highest number of accidents. This trend is particularly strong in California, having approximately double the number of accidents compared to any other state. Other highly populated states, including Florida, Texas, New York, and the Carolinas, also experience many accidents. These plots conveyed the importance of population and location in accidents and made geography another central focus of our final visualizations.

We also explored weather conditions as one of the initial directions of analysis to identify any noticeable patterns relating to the variable. A waffle chart in Figure 10 was used to visualize the distribution of accidents across grouped weather categories, while a normalized stacked bar in Figure 11 showed a comparison of severity levels in each. The visualizations suggested that most accidents occurred during clear and cloudy conditions, with only moderate variation in severity across the categories. However, without knowing the baseline proportion of days with each weather condition, it was difficult to draw meaningful conclusions from accident counts alone. In addition, the overlap between geographic factors and weather made it difficult to isolate weather as a singular influence on accident trends, so the variable was not included as an investigative direction for explanatory visualizations.

Our final direction of exploration was severity in accidents. The severity of accidents was measured on the disruption in traffic flow, ranging from 1, minor delays, to 4, major delays. A bar chart was used to display the proportions of accidents falling into each category, shown in Figure 3 of Appendix B. The bar chart of accident count by severity level shows that a vast majority accidents that occur are considered severity level 2 or 3. Few accidents are level 1, or minor, and few reach the threshold of high severity at level 4. Given the practical importance of severity as a measure accident impact, this was the final direction we chose to investigate. Together, our exploratory analysis helped us choose geographic, temporal, and severity patterns as the most important and interpretable aspects of the data, which our final visualizations were based upon.

Visualizations:

The first direction we chose to pursue for our final visualizations was geographic. In our exploratory analysis, we identified that the population appeared to be a driving factor of the raw accident count. However, we wanted to explore this further and see if these states truly had more accidents than others. Our first visualization, seen in Figure 4, is two choropleths that combine to reveal which states have disproportionately high or low accident rates relative to their population size using color encoding.

The top choropleth displays accidents per 100,000 residents by state, normalizing for population. This allows for a fair comparison of accidents in each state, regardless of size. Normalization was vital, as it showed trends our exploratory choropleth did not. South Carolina, California, Oregon, and Florida emerged as the states with the highest accident rates, while Maine and Vermont were the lowest.

The lower choropleth displays the Bayesian surprise in each state's accident rate. Bayesian surprise is a measure of how much each state's observed accident rate deviates from what its population predicts. The red states had significantly more accidents than expected, while the blue states had fewer than expected. South Carolina had by far the highest surprise value, meaning far more accidents occur than its population would predict. This suggests factors beyond population are influencing accidents in the state. Most states fell slightly below their expectation, with Maine and Vermont being the lowest. Together, these choropleths show where accidents are more common than expected and where driving may be more dangerous, going beyond the initial choropleth's population-based trend.

Our second geographic visualization is another set of choropleths, conveying the percentage of traffic accidents categorized as high severity by state. Severity levels 3 and 4 were combined to represent high severity, and the results can be seen in Figure 5. The top choropleth shows the proportion of high-severity accidents by state. Georgia and Rhode Island stand out at the top of the scale, with nearly 50% of their accidents classified as high severity. In contrast, Montana and North Dakota had the lowest proportion of accidents recorded. This map is especially interesting when considered with the normalized traffic accident rates choropleths from figure 4. States with high accident rates, like South Carolina, do not necessarily have many severe accidents. This implies that frequency and severity are influenced by different factors.

The bottom choropleth again uses Bayesian surprise to convey which states have unexpected proportions of high severity accidents. The Great Lakes region, the Northeast, Texas, and Georgia all displayed higher proportions of high severity accidents than expected. Western states and the Great Plains region generally fell below expected proportions. The

regionality seen suggests that aspects such as traffic density, infrastructure, and concentration of urban centers may influence accident severity. Together, the geographic visualizations reveal that accident frequency and severity follow distinct regional patterns with different underlying drivers.

The next investigative direction we pursued was temporal. In our exploratory analysis, there were obvious accident patterns based on the time of day and day of the week. The first visualization we used to explore this was a heatmap, seen in Figure 6. The heatmap was constructed to show the distribution of accidents by hour of the day, seen on the y-axis following a calendar-view, and day of the week, seen on the x-axis going from Monday to Sunday. The color scale of the heatmap is plasma, yellow indicating the maximum number of accidents (7,716) and purple indicating the minimum number of accidents (655).

The heatmap confirms and furthers the findings from exploratory histograms. Accidents are heavily concentrated on weekdays during rush hour, from 7AM – 9AM and 4PM – 6PM. Interestingly, the most accidents occur between 7AM – 8AM on Wednesday. Nights, from 8PM – 4AM, and weekends have much lower concentrations of accidents. The heatmap combines our exploratory findings, revealing how the daily and hourly temporal patterns combine to form a clear trend in accidents.

Our second temporal visualization were radial plots of the accident counts by month in the Northern and Southern halves of the U.S. The blue color scale ranges from a minimum of 7,170 to a maximum of 22,541 accidents. Both radial plots show the highest number of accidents that occur in December and the lowest occur in March. The sections of the Southern States' radial plot are much longer than the Northern States' corresponding sections indicating most of the reported accidents occur in the Southern states.

The final investigative direction we chose to pursue was accident severity. As previously mentioned, severity was categorized in four levels of traffic impact, with level 1 being the lowest and level 4 being the highest. In our exploratory analysis, we discovered that the majority of accidents are categorized as level 2, followed by level 3, and few were labeled level 1 or 4. Mosaic plots were used to further explore the severity trends in the data.

The first mosaic plot, seen in Figure 8, shows the distribution of accident severity in a 24-hour time frame. Each column represents a 3-hour time period, with the total number of accidents per time frame encoded with width, the proportion of accident severity encoded with length, the total number of accidents per severity level per time frame is encoded with area. The mosaic plot's encoding allows both total accidents and severity to be conveyed. The visualization reveals a significant difference in accidents counts during the day and at night. Daytime accidents, particularly during rush hour periods, accounted for majority of accidents. In addition, a vast majority of these accidents were severity level

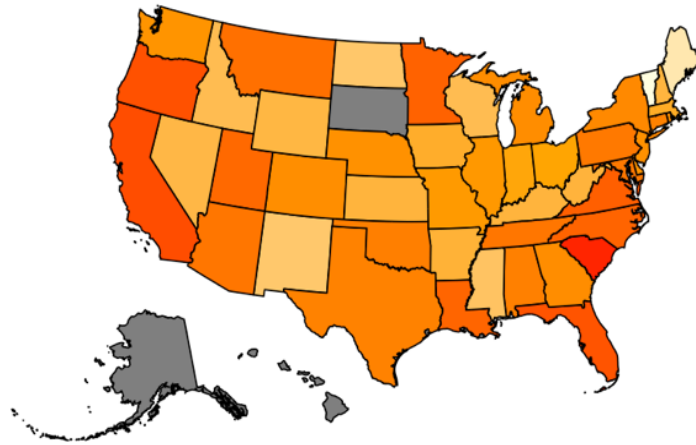
2. In contrast, at night far fewer total accidents occurred, but they were more likely to be severe. Level 4 severity made up a much larger portion of the accidents at night, particularly between 9PM – 3AM. This suggests that although nighttime driving results in less total accidents, these accidents are much more likely to be severe.

The second mosaic plot, seen in Figure 9, was generated to show the total number of accidents that occurred per year with R's `mosaicplot()` function. Severity is encoded with color, with green, yellow, orange, and red representing levels 1-4, respectively. Like the previous mosaic plot, total number of annual accidents is encoded with width, the proportion of accident severity is encoded with length, and area encodes the total number of accidents per severity level per year.

This plot reveals important insight into the evolution of car accidents in the U.S since 2016. First, accidents totals have increased annually since 2017, seen in the widening of the columns until 2022 (2023 not fully sampled). Second, the proportion of high-severity accidents has steadily declined since 2018, with much less orange and red in the plot. This visualization expands on the temporal and severity insights of our project, displaying the progression of traffic accidents.

US Traffic Accidents

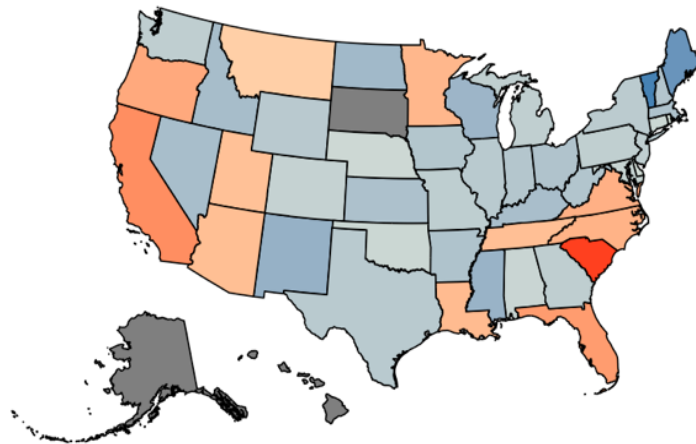
Per 100,000 Residents



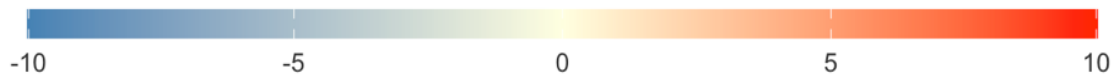
Accidents



Bayesian Surprise in Accident Rates



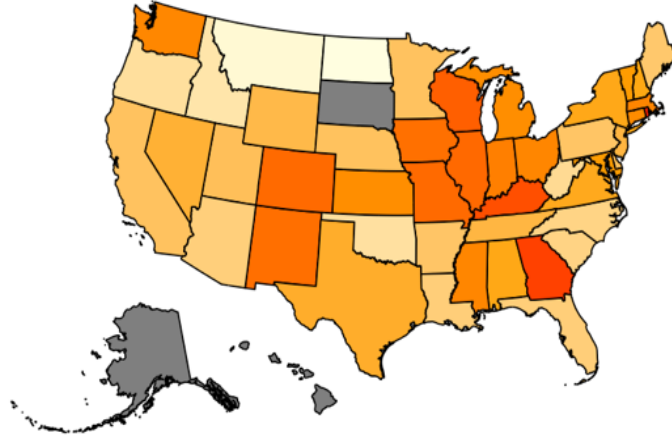
Bayesian Surprise



*Grey indicates sparse coverage

Figure 4 – Rate of US traffic accidents and Bayesian surprise choropleths

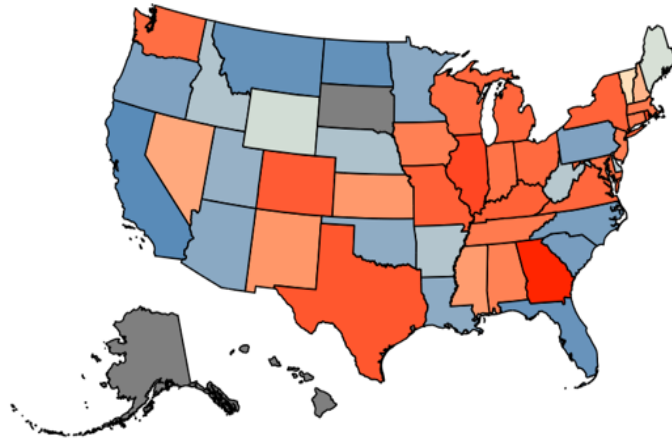
Percentage of High Severity Traffic Accidents



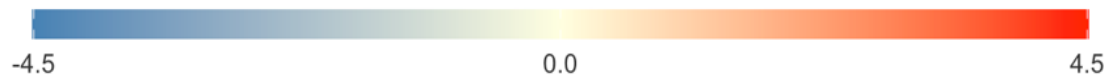
Proportion severity 3 or 4



Bayesian Surprise in Accident Severity

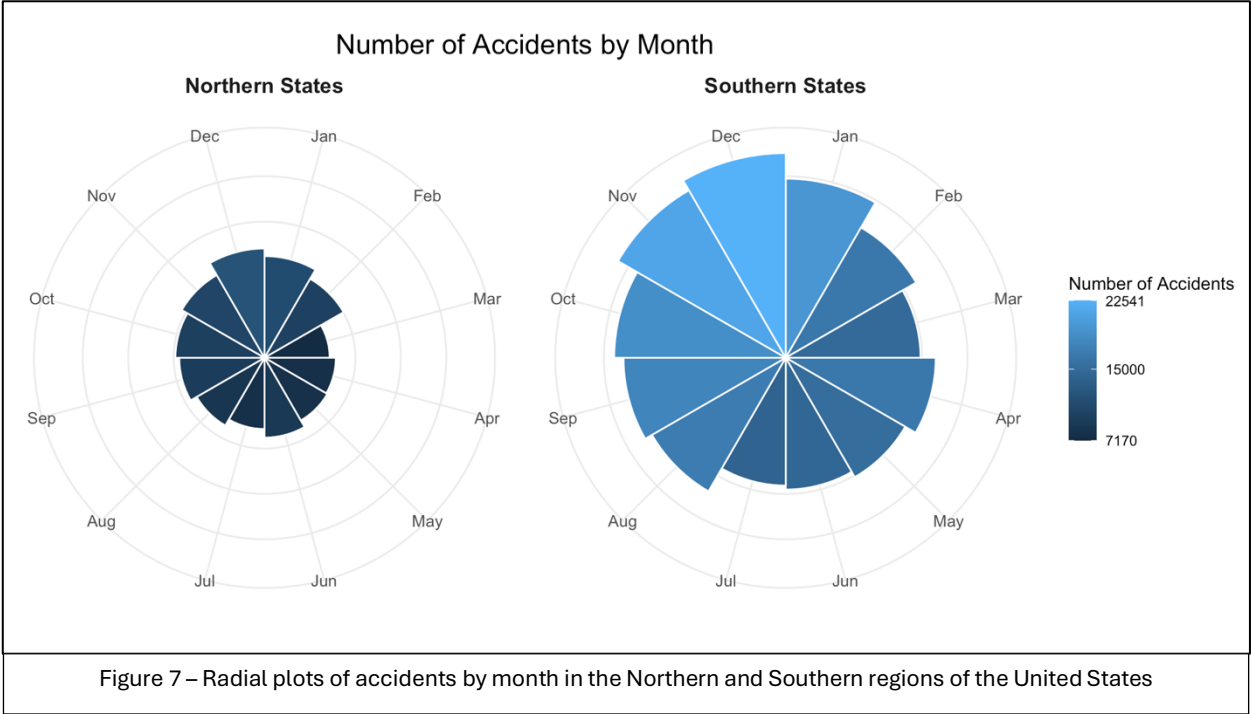
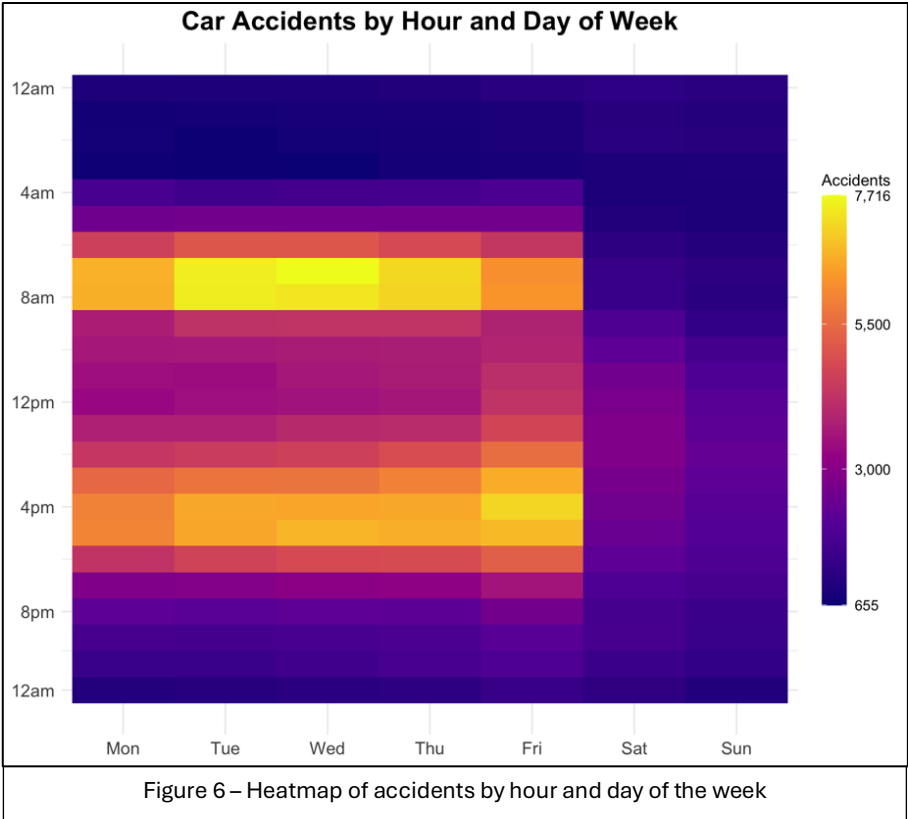


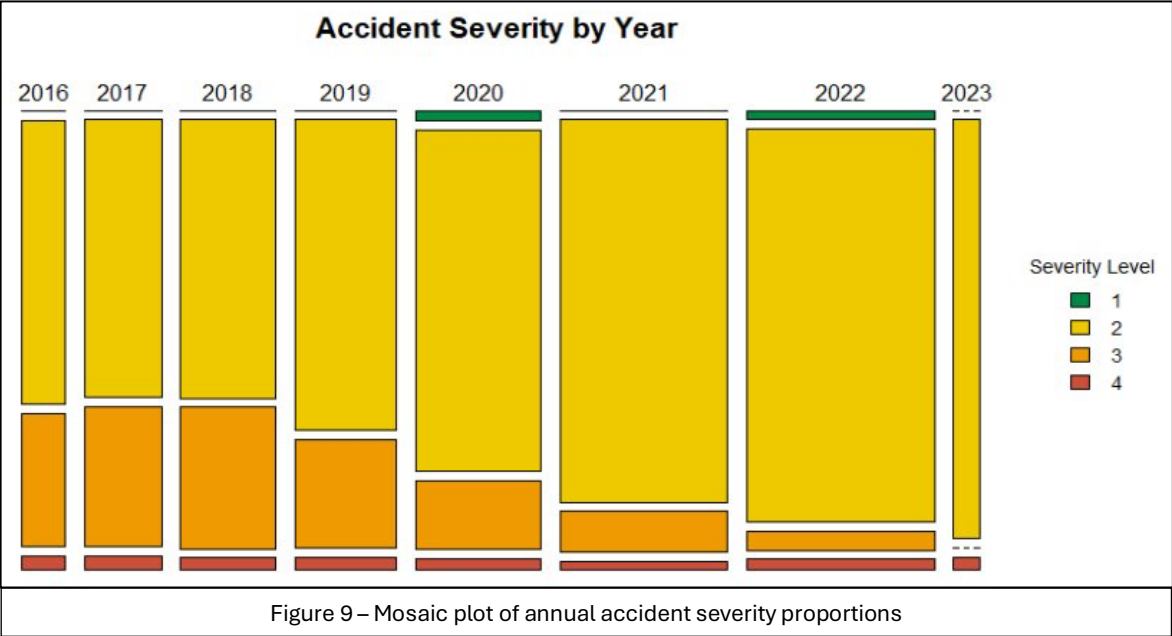
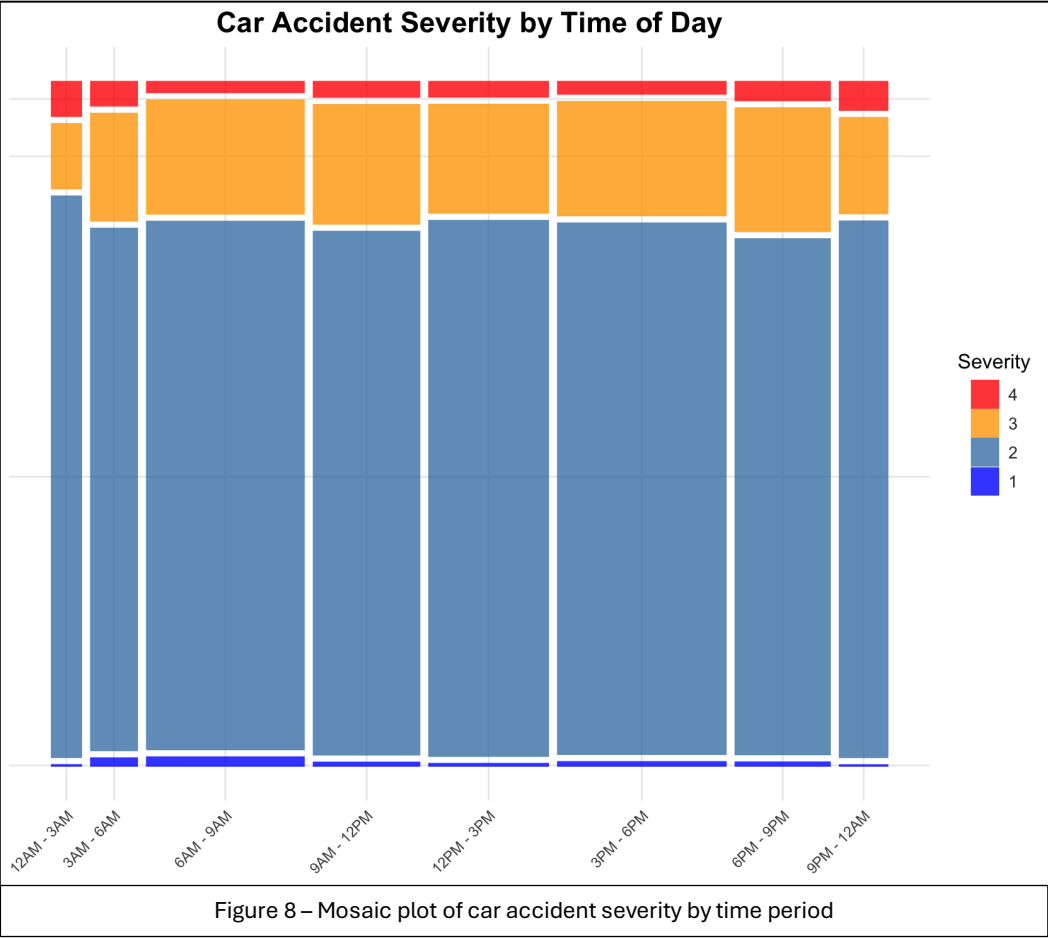
Bayesian Surprise



*Grey indicates sparse coverage

Figure 5 – Percentage of high severity traffic accidents and Bayesian surprise choropleths





Analysis and Discussion:

Our analysis of the U.S. car accidents dataset revealed several meaningful patterns across geographic, temporal, and severity dimensions. By expanding on our exploratory visualizations and analysis, we were able to discover and convey insightful trends about where, when, and how accidents occur through our explanatory visualizations.

The geographic visualizations, our choropleths, showed that raw accident counts seem to follow population closely, but normalization revealed more interesting trends. South Carolina, California, and Oregon emerged as states with disproportionately high accident rates relative to their population, while Vermont and Maine were disproportionately low. However, this did not correlate to high severity. Georgia and Rhode Island showed the highest percentage of severe accidents, and California and South Carolina were relatively average, despite their rate of accidents. This discrepancy suggests that accident frequency and severity are influenced by different factors and need to be addressed independently.

Our temporal visualizations revealed clear patterns about when accidents are most likely to occur. The heatmap shows that accidents occur most frequently during weekdays, peaking during morning and evening rush hours. The 7AM – 8AM slot stands out as the period with the highest number of accidents from Monday to Friday. There is also a spike on Friday afternoon, likely due to higher traffic at the end of the week. Far fewer accidents occur overnight and on weekends. This suggests that commuting behavior is a major driver of accident frequency.

Examining pattern at a seasonal level, the radial plot shows the total number of accidents per month for northern and southern states. The southern states recorded substantially more accidents overall, which may partly reflect differences in population size and road network density. Both regions show a seasonal variation in the number of accidents, with accident counts tending to be higher in winter months (November–January). This might be due to increased holiday travel and adverse weather conditions. The northern states show a more uniform distribution across months, suggesting that other regional factors, such as infrastructure, driving behavior, or reporting practices, may also influence accident frequency.

Our severity visualizations revealed important patterns in both the distribution and evolution of accident severity. The first mosaic plot shows that severity level 2 accidents dominate across all time slots. Severity level 1 accidents are nearly absent throughout the day, suggesting that very minor incidents are rarely reported. The 12AM–6AM bars are visibly narrower, which confirms that fewer accidents occur overnight, however the proportion of

the most severe level 4 accidents is higher during these hours, possibly due to driver fatigue, higher speeds on emptier roads, or slower emergency response times. The wider bars during the 6AM–9AM and 3PM–6PM windows reflect the higher volume of accidents during morning and evening commutes.

The second mosaic plot shows that nearly 80% of recorded accidents in the US from 2016-2023 are severity 2, meaning that the impact on traffic flow is normally minor. The proportion of high-severity accidents (levels 3 and 4) has declined since 2018, which may reflect safer vehicles and roads, improved first responder time, or changes in data collection practices. 2020 shows a visibly narrower bar, which is consistent with reduced traffic during the COVID-19 pandemic. Accident volume increased in 2021 and 2022, likely as traffic returned to normal following the pandemic. The 2023 bar appears narrow simply because the dataset does not cover the entire year, so it does not reflect a real decrease in accidents.

Conclusion:

Our visualizations revealed that accident risk in the United States follows predictable geographic, temporal, and seasonal patterns, suggesting several possible opportunities for targeted intervention. Focused driving and awareness campaigns, law enforcement, and emergency-response staffing during peak traffic hours can alleviate some accident risk. Nighttime safety measures should be improved to decrease severe crashes. Winter months should also receive additional attention, as accident numbers tend to rise during this period, potentially due to adverse weather conditions and increased holiday travel. Southern states, which recorded significantly more accidents overall, may benefit from greater resource allocation and road safety investment. States with higher proportions of severe crashes should be prioritized with targeted interventions as well. Further analysis, such as regression modeling or clustering of states with similar risk profiles, and detailed weather condition analysis and impact, could provide additional insight in the future. Overall, our findings suggest that a combination of time-based, seasonal, and regional strategies would be most effective in reducing both the frequency and severity of motor vehicle accidents across the United States.

Appendix A – Individual Reports:

Kira Damo:

I contributed to the project by creating visualizations during the exploratory phase and although they were not used in the final production of the report, it was an insightful opportunity of making decisions based on data, audience, and message. The visualizations that I contributed were a waffle chart of the environmental factors in the second milestone and an animated area graph of accidents by hour over months in the third milestone.

The only binary variables in the graph were environmental factors such as stop signs and traffic signals. I was personally interested in seeing if these factors contributed to the severity and number of accidents. To create the waffle chart, I created a heat map with borders that formed a 10 x 10 grid. The tiles were filled as the percentage of the data where each environmental factor was counted as true. Eventually I wanted to color code severity but after seeing the effectiveness of other visualizations and after discussing the intentions of our message, these environmental factors were not relevant to the themes of time and geography.

The animated graph was discarded after the third milestone. The start time was reformatted using the Lubridate package. The number of accidents were counted for each hour by month and sorted. To portray a relationship of accidents over time, I animated an area graph. This graph did not show a substantial trend and was not nearly as effective as the other visualizations. Even if more refining was done to the transition parameters and aesthetics, the graph would be deemed unnecessary to the investigations and difficult to decode to an extremely broad audience of American drivers.

My most prominent contribution was the analysis of visualizations and presenting the underlying messages in the final draft. I presented the heatmap of car accidents by hour and day of the week and the mosaic plot that showed car accident severity by time of day. Understanding data visualizations requires an element of reading between the lines and gathering context.

I drew many conclusions on the visualizations from the powerful channel of personal experience! I found this dataset interesting because there is an element of humanism that allows more underlying factors to shine through. Having the experience of sitting in traffic during rush hour, seeing people on their phones in the car, witnessing accidents at night are experiences that are common and enhances the connection with the audience.

The curriculum of the class equipped me with a plethora of options to convey findings of any kind of dataset. Tailoring different channels and choosing a collection of visualizations to cater to the data, audience, and message is what builds the integrity of data

scientists and this project gave my group the chance to apply those principles. Although there is ground truth in algorithms and computations, there are subjective nuances and stories that data visualizations can convey.

Massiel Payero Garcia:

I assisted in getting people together to form a group. I created a Discord server to make communicating with the group easier. I participated in looking for different datasets that the group could use for the project. I proposed three datasets for consideration. I helped create visuals for the exploratory analysis phase. I created a mosaic plot that showed the relationship between accident severity and year. I also created a choropleth that showed the percentage of severity 4 accidents by state. There wasn't much data for some of the states. This skewed some of the results. We ended up changing this visualization for the project for this reason. I also made histograms that explored the number of accidents by hour of day and day of the week. The visualizations I created in the exploratory phase helped the group choose a direction to take the final project.

In addition to contributing my own visualizations to the project, I also advised other group members about edits that they could make to their visualizations. For example, I suggested greying out states with insufficient data for the choropleths. This helped to visually separate excluded states from states with low accident proportions/counts. Other group members also helped to make my visualizations better. For example, Alexander made several improvements to my initial choropleth. It was useful to hear different perspectives. I believe I was proactive in contributing my portions of the milestones and finished my work in a timely manner.

While working on this project, I learned that there can be hidden information that can be discovered through visualizations. Some of the visualizations created for this project reveal trends in the data that would not have been apparent otherwise. This reinforced the importance of creating visualizations during the exploratory analysis phase. The use of color in particular is something that I had not given much attention to in the past, but I understand now what a large difference it can make to the information communicated by a visualization. As a data science student, I know that the skills I have learned from this class will become useful in my future work. I expect to make many data visualizations in the future.

Katrina Grigorovica:

I contributed to the project by helping to evaluate potential datasets during the initial planning stage and participating in discussions about visualization selection, interpretation, and development of the final narrative. I also assisted in structuring portions of the report, particularly the organization of the exploratory analysis. Most of my contributions occurred during the exploratory analysis phase, where I primarily focused on examining the weather condition variable and its relationship to accident frequency and severity within the dataset.

To investigate this relationship, I analyzed how accidents were distributed across various weather conditions and whether certain conditions were associated with higher severity levels. Since the dataset contained a significant number of highly specific condition labels, I grouped similar conditions into broader categories to improve clarity. Together, the visualizations provided the group with an insight into the relationship between weather conditions and accident outcomes and helped determine if the variable showed meaningful patterns worthy of deeper analysis.

While some noticeable patterns were discovered within the variable, after careful consideration together with the group members, the findings ultimately were not included in the final analysis. Relative to the findings in temporal and geographic visualizations, these patterns were not strong and consistent enough to support clear interpretation. Even though the visualizations were not included in the final presentation, the exploratory work was valuable to help narrow down the project's final narrative.

In addition to the exploratory analysis, I also contributed to reviewing and refining components of the presentation materials. Like other members of the group, I assisted with the discussions regarding the interpretation of results both in the presentation and the report. This project gave me hands-on experience working with a large dataset, and I gained a better understanding of how significant the role of visualization is not only in presenting the results but also in exploring data. Investigating a less conclusive relationship taught me that every stage is valuable for understanding the data and shaping the narrative. Overall, the project strengthened both my critical thinking and technical skills, as well as increased my awareness of careful examinations before presenting data and conclusions.

Zofia Landowska:

My first contribution to the project was setting up the GitHub repository, which kept our work organized and allowed us to collaborate on code without conflicts. In the exploratory analysis phase, I focused on exploring two areas: the density of accidents across regions within the states with the highest total accident counts and investigating how road features, such as traffic signals and junctions, relate to accident frequency. For the density analysis, I found that although California has the highest national total, accidents are spread across many cities. Florida's accidents are heavily concentrated in the Miami area, which recorded more accidents than any single city in California. Despite this, both distributions largely followed population density, which limited how informative these maps were on their own. The road feature analysis did not produce meaningful findings, as the selected features showed little variation in relation to accident frequency. While neither of my exploratory visualizations was included in the final report, they helped the team rule out infrastructure-based explanations and focus on temporal and geographic trends instead.

My final visualization was the radial plots showing monthly accident counts for northern and southern states. I chose a radial format because this layout better communicates the cyclical nature of monthly data. I split the plots by region to test whether climate differences would produce distinct seasonal patterns. A key design decision was ensuring both plots used the same scale, since mismatched scales can make one region appear smaller than the other even when the difference is meaningful. I also contributed to writing the final report sections and interpreting visualizations produced by other team members.

This project taught me a lot about data visualization in practice. I learned that the choice of chart type directly shapes the message of the visualization. For example, radial plots made seasonal cycles immediately readable in a way a standard bar chart would not have. Heatmaps were more effective than scatter plots for spotting patterns across two categorical dimensions. I also learned that visualizations can be misleading if not carefully designed. If two connected charts use different scales, the graphical representation no longer accurately reflects the data, which can mislead readers. An unexpected lesson came from the exploratory phase because none of my work was used in the final report. Finding that accident density simply tracked population density, and that road features showed no clear signal, was still valuable because it narrowed the project's focus. Overall, this project showed me that good data visualization is not just about making charts look appealing but about communicating findings clearly and honestly to the reader.

Alexander Williams:

I believe our group collaborated well together, with everyone providing significant contributions. I was the liaison of the group, so throughout the project I was responsible for submitting assignments in a timely manner, meeting with Dr. Brown for feedback and progress updates, and final edits prior to submission. I also contributed to coordinating group communication and decision making on visualization selections.

Initially after Milestones 0+1 of group formation, all group members, including myself, found several possible datasets to use for the projects. Ultimately one of my datasets, the *US Accidents (2016 - 2023)*, was used in the project. Milestone 2 was the exploratory analysis phase of the project, and I contributed several visualizations here. My visualizations included a bar chart of accidents by day of the week, showing that far more accidents occur on weekdays than weekends, a bar chart of accident severity, revealing that a vast majority of accidents are severity level 2 or 3, a bar chart of states with the most accidents, and a choropleth of accident counts by state. All these visuals were included in the submission and were part of the basis for our investigative directions, geographic, temporal, and severity. I also researched and included two of the related visualizations, the car accident fatality choropleth and the accident time series plot.

Milestone 3 included some early drafts of our group's visualizations and the solidification of our investigative directions. In this assignment, I described the directions we were planning to pursue and included two of my initial explanatory visualizations. I created the mosaic plot of accident severity by time of day, exploring severity trends in the data, and the heat map of car accidents by hour and day of the week, exploring temporal trends in the data. I was also the representative group member to meet with Dr. Brown and discuss these visualizations to get feedback and improve them.

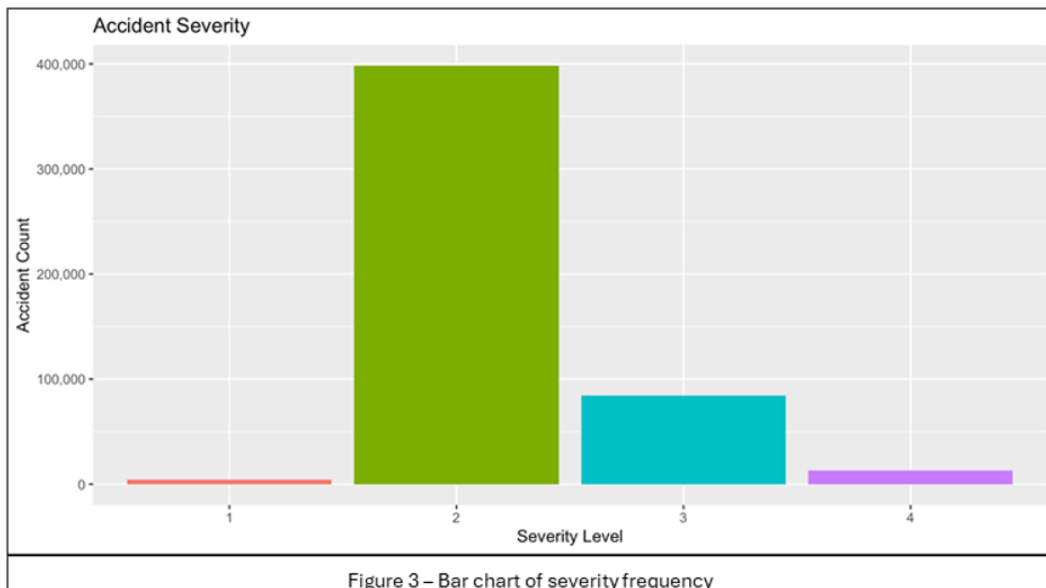
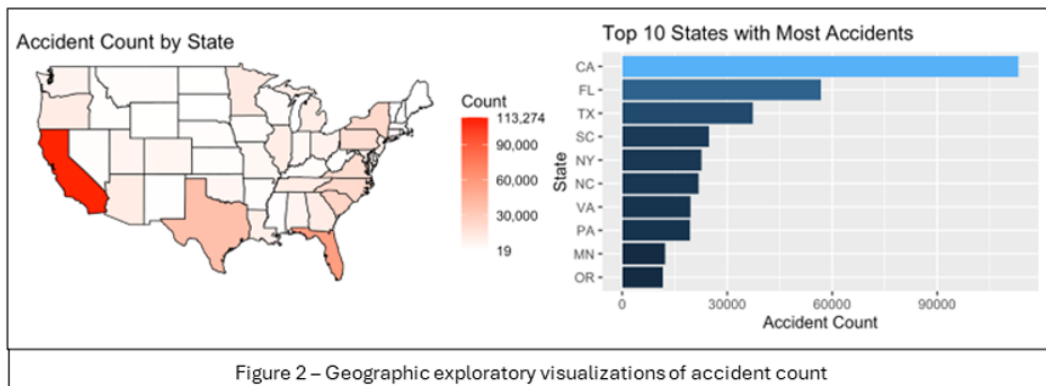
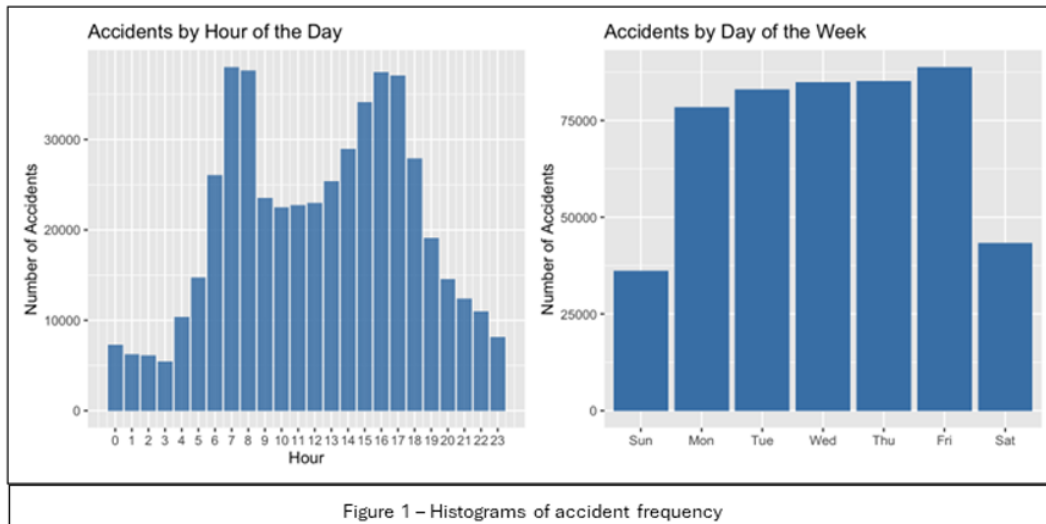
Milestone 4 was our presentation, with all the final drafts of our explanatory visualizations included. I created the slides and structure for the presentation, and several of my visualizations were included. For our geographic visualizations, I created the choropleth of U.S. traffic accidents normalized and both Bayesian surprise choropleths. After my meeting with Dr. Brown, I explored the Bayesian surprise resource provided and used it to create the visualization components, which I believe gave the plots far more explanatory power and insight. Next, for temporal visualizations, my revised heat map of car accidents by hour and day of the week from Milestone 3 was included. Finally, for further severity investigation, my mosaic plot of car accident severity by time of day from Milestone 3 was revised and included. I contributed to the key findings and practical insight slides as

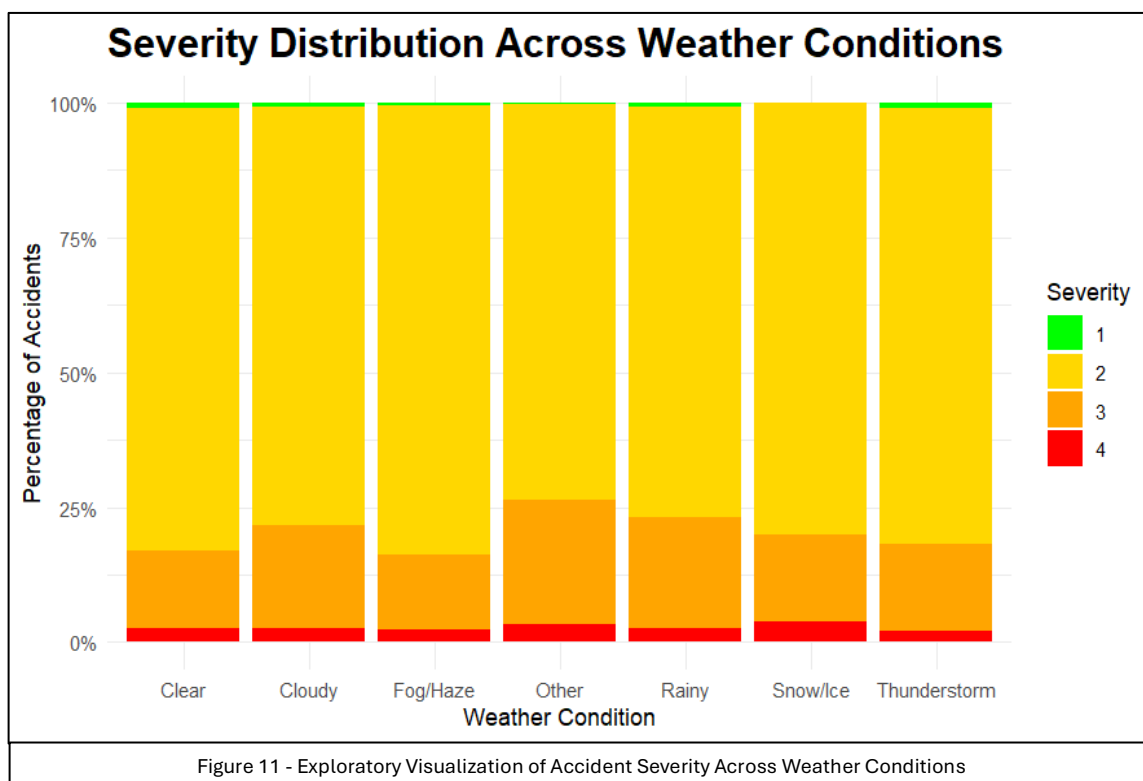
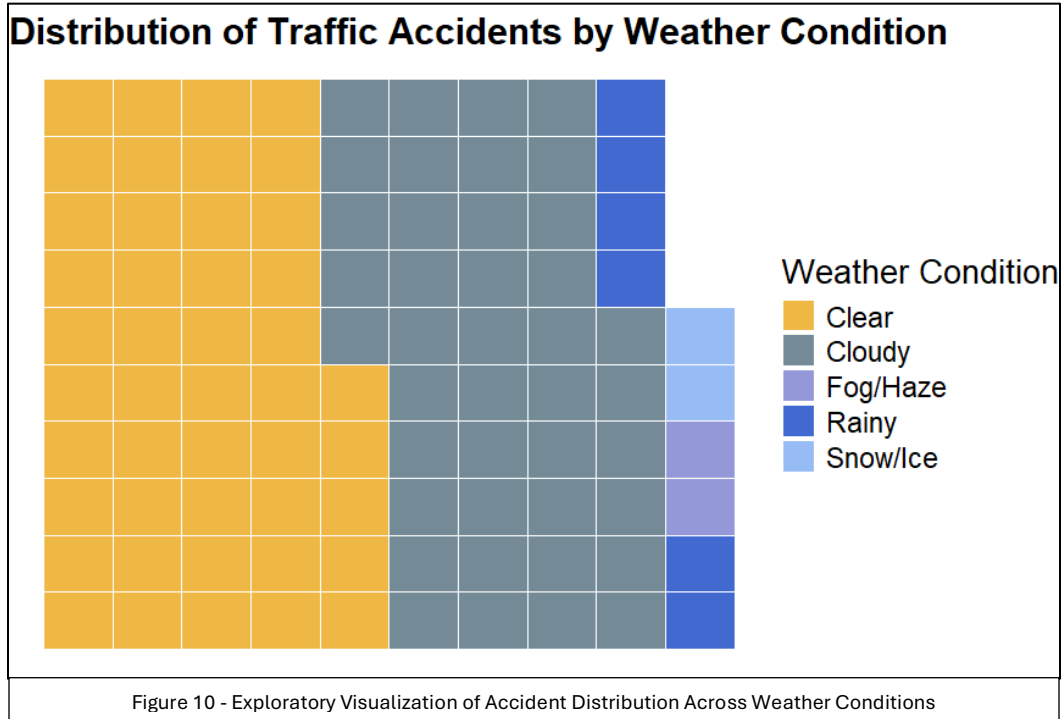
well. During the presentation, I covered the introduction slides and the first choropleth visualization. It is worth noting that many visualizations in various investigative directions were created by all group members, but we agreed that the ones selected provided the best insight.

Lastly, in the final report I wrote the introduction, explained my exploratory visualizations in Figures 1, 2, and 3, analyzed all of the choropleth final visualizations (Figures 4 and 5) and car accident severity by time-of-day mosaic plot (Figure 8), contributed to the analysis of my temporal heatmap visualization (Figure 6), helped write the discussion and conclusion, and provided code for all of my visualizations included.

This project taught me multiple things about visualization. First, it taught me the value of explanatory visualizations. As a data scientist student, a majority of the visualizations I produce are exploratory for machine learning. This was my first hands-on experience making explanatory visualizations, and it showed how valuable they can be. The final visualizations in our project were far more complex than our initial ones but were also easier to read and convey deeper meaning. An appropriate explanatory visualization can present data in a neat, digestible manner. Next, I learned that a combination of visualizations can be used to make a more comprehensive story. The four choropleth visualizations in Figures 3 and 4 are a good example of this. The combination of accidents rates, percentage of high severity accidents, and Bayesian surprise for both, allowed us to see trends that were not visible in the exploratory choropleth or in either plot alone. Insight can be cumulative with visualizations. Finally, I learned the importance of normalization and fair representation in visualization. Again, relating to the choropleths, the normalization of the data completely changed the message of the data. At the beginning of our analysis, the only relevant factor in accidents appeared to be population. After normalization, we were able to explore several different and more interesting trends in the data. This project changed how I view the communication of data and improved my understanding of the utility in a good visualization.

Appendix B – Exploratory Visualizations:





Appendix C – Important Code:

Full Code:

<https://github.com/ZofiaLandowska/Data-Vis-Final-Project.git>

Figure 1 Code:

```
# accidents by day of the week
accidents %>%
  filter(!is.na(DayOfWeek)) %>%
  count(DayOfWeek) %>%
  ggplot(aes(x=DayOfWeek, y=n)) +
  geom_col(fill="steelblue") +
  labs(title = "Accidents by Day of the Week",
       x = NULL,
       y = "Number of Accidents") +
  theme(legend.position = "none")
```

Figure 2 Code:

```
# accidents by hour of the day
accidents %>%
  count(Hour) %>%
  ggplot(aes(x=Hour, y=n)) +
  geom_col(fill="steelblue") +
  scale_x_continuous(breaks=0:23) +
  labs(title = "Accidents by Hour of the Day",
       x = "Hour",
       y = "Number of Accidents")
```

```
# aggregation of states
state_counts <- accidents %>%
  count(State) %>%
  mutate(region = tolower(state.name[match(State, state.abb)]))

# choropleth of accident frequency
map_data("state") %>%
  left_join(state_counts, by="region") %>%
  ggplot(aes(x=long, y=lat, group=group, fill=n)) +
  geom_polygon(color="black", linewidth=0.3) +
  scale_fill_gradient(low="white", high="red",
                    labels=scales::comma,
                    breaks=c(min(state_counts$n), 30000, 60000,
                             90000, max(state_counts$n))) +
  coord_fixed(1.3) +
  labs(title = "Accident Count by State", fill = "Count") +
  theme_void()
```

Figure 3 Code:

```
ggplot(accidents, aes(x=Severity, fill=Severity)) +
  geom_bar() +
  scale_y_continuous(labels = scales::comma) +
  labs(title = "Accident Severity",
       x = "Severity Level",
       y = "Accident Count") +
  theme(legend.position = "none")
```

Figure 4 Code:

```
# accident rate choropleth
ggplot(allStates) +
  geom_sf(aes(fill=accidents_per_100k, color="black", linewidth=0.3) +
    scale_fill_gradientn(colors=c("lightyellow", "orange", "red"),
      trans="log10",
      limits=c(rate_min, rate_max),
      breaks=c(rate_min, 15, 25, 50, 100, 200, rate_max),
      labels=~comma(round(.x)),
      na.value="grey50",
      name="Accidents",
      guide=guide_colorbar(direction="horizontal",
        title.position="top",
        title.hjust=0.5,
        barwidth=unit(15, "cm"),
        barheight=unit(0.4, "cm")))) +
  labs(title = "US Traffic Accidents",
    subtitle = "Per 100,000 Residents",
    caption = paste0("*Grey indicates sparse coverage")) +
  theme_void() +
  theme(plot.title = element_text(size=16, face="bold", hjust=0.5),
    plot.subtitle = element_text(size=12, color="black", hjust=0.5),
    plot.caption = element_text(size=10, color="black", hjust=0.75),
    legend.position = "bottom",
    legend.title = element_text(size=12, face="bold", hjust=0.5),
    legend.text = element_text(size=10, color="grey20"),
    legend.margin = margin(t=5),
    plot.margin = margin(10, 10, 10, 10))
```

```
# Bayesian surprise choropleth
max_abs <- max(abs(allStates$surprise), na.rm=TRUE)

ggplot(allStates) +
  geom_sf(aes(fill=surprise, color="black", linewidth=0.3) +
    scale_fill_gradient2(low = "steelblue",
      mid = "lightyellow",
      high = "red",
      midpoint = 0,
      limits = c(-max_abs, max_abs),
      breaks = c(-max_abs, -5, 0, 5, max_abs),
      labels = ~comma(round(.x)),
      na.value = "grey50",
      name = "Bayesian Surprise",
      guide = guide_colorbar(direction="horizontal",
        title.position="top",
        title.hjust=0.5,
        barwidth=unit(15, "cm"),
        barheight=unit(0.4, "cm")))) +
  labs(title = "Bayesian Surprise in Accident Rates",
    caption = "*Grey indicates sparse coverage") +
  theme_void() +
  theme(plot.title = element_text(size=16, face="bold", hjust=0.5),
    plot.subtitle = element_text(size=12, color="black", hjust=0.5),
    plot.caption = element_text(size=10, color="black", hjust=1),
    legend.position = "bottom",
    legend.title = element_text(size=12, face="bold", hjust=0.5),
    legend.text = element_text(size=10, color="grey20"),
    legend.margin = margin(t=5),
    plot.margin = margin(10, 10, 10, 10))
```

Figure 5 Code:

```
# high severity map
ggplot(allStates_sev) +
  geom_sf(aes(fill=prop_severe), color="black", linewidth=0.3) +
  scale_fill_gradientn(colors = c("lightyellow", "orange", "red"),
    limits = c(sev_min, sev_max),
    breaks = c(sev_min, 0.6, 0.7, 0.8, sev_max),
    labels = ~percent(.x, accuracy=1),
    na.value = "grey50",
    name = "Proportion severity 3 or 4",
    guide = guide_colorbar(direction="horizontal",
      title.position="top",
      title.hjust=0.5,
      barwidth=unit(15, "cm"),
      barheight=unit(0.4, "cm")))) +
  labs(title = "Percentage of High Severity Traffic Accidents",
    caption = "**Grey indicates sparse coverage") +
  theme_void() +
  theme(plot.title = element_text(size=16, face="bold", hjust=0.5),
    plot.subtitle = element_text(size=12, color="black", hjust=0.5),
    plot.caption = element_text(size=10, color="black", hjust=1),
    legend.position = "bottom",
    legend.title = element_text(size=12, face="bold", hjust=0.5),
    legend.text = element_text(size=10, color="grey20"),
    legend.margin = margin(t=5),
    plot.margin = margin(10, 10, 10, 10))
```

```
# Bayesian Surprise severity map
ggplot(allStates_sev) +
  geom_sf(aes(fill=surprise), color="black", linewidth=0.3) +
  scale_fill_gradient2(low = "steelblue",
    mid = "lightyellow",
    high = "red",
    midpoint = 0,
    limits = c(-sev_abs, sev_abs),
    breaks = c(-sev_abs, 0, sev_abs),
    labels = ~comma(round(.x, 1)),
    na.value = "grey50",
    name = "Bayesian Surprise",
    guide = guide_colorbar(direction = "horizontal",
      title.position = "top",
      title.hjust = 0.5,
      barwidth = unit(15, "cm"),
      barheight = unit(0.4, "cm")))) +
  labs(title = "Bayesian Surprise in Accident Severity",
    caption = "**Grey indicates sparse coverage") +
  theme_void() +
  theme(plot.title = element_text(size=16, face="bold", hjust=0.5),
    plot.subtitle = element_text(size=12, color="black", hjust=0.5),
    plot.caption = element_text(size=10, color="black", hjust=1),
    legend.position = "bottom",
    legend.title = element_text(size=12, face="bold", hjust=0.5),
    legend.text = element_text(size=10, color="grey20"),
    legend.margin = margin(t=5),
    plot.margin = margin(10, 10, 10, 10))
```

Figure 6 Code:

```
# aggregation
hourday_acc <- accidents %>%
  group_by(DayOfWeek, Hour) %>%
  summarise(crashes=n(), .groups="drop") %>%
  mutate(DayOfWeek=factor(DayOfWeek, levels= c("Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun")))

# heatmap
ggplot(hourday_acc, aes(x=DayOfWeek, y=Hour, fill=crashes)) +
  geom_tile() +
  scale_fill_viridis_c(option = "plasma",
                       name = "Accidents",
                       breaks = c(min(hourday_acc$crashes), 3000, 5500, max(hourday_acc$crashes)),
                       labels = ~comma(round(.x))) +
  scale_y_continuous(breaks = c(0, 4, 8, 12, 16, 20, 23),
                    labels = c("12am", "4am", "8am", "12pm", "4pm", "8pm", "12am"),
                    trans = "reverse") +
  labs(title="Car Accidents by Hour and Day of Week") +
  theme_minimal() +
  theme(plot.title = element_text(size = 18, face="bold", hjust = 0.5),
        axis.text = element_text(size = 12),
        legend.text = element_text(size = 10),
        axis.title = element_blank(),
        legend.key.height = unit(2, "cm"))
```

Figure 7 Code:

```
south_data <- accidents %>%
  filter(State %in% southern_states) %>%
  group_by(Month) %>%
  summarize(count = n()) %>%
  mutate(Region = "South")

north_data <- accidents %>%
  filter(State %in% northern_states) %>%
  group_by(Month) %>%
  summarize(count = n()) %>%
  mutate(Region = "North")

combined_data <- bind_rows(south_data, north_data)

# Calculate breaks from combined data
min_count <- min(combined_data$count)
max_count <- max(combined_data$count)
mid_count <- round(mean(c(min_count, max_count)), -3)
breaks_vals <- c(min_count, mid_count, max_count)

# Shared y-axis ceiling
y_max <- max(combined_data$count)

# Plot
ggplot(combined_data, aes(x = Month, y = count, fill = count)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  scale_fill_gradient(
    low = "#132B43", high = "#56B1F7",
    breaks = breaks_vals,
    labels = c(min_count, mid_count, max_count)
  ) +
  scale_y_continuous(limits = c(0, y_max)) +
  coord_polar() +
  facet_wrap(~ Region) +
  theme_minimal() +
  theme(
    axis.text.y = element_blank(),
    plot.title = element_text(size = 17, hjust = 0.5),
    axis.text = element_text(size = 10),
    strip.text = element_text(size = 13, face = "bold")
  ) +
  labs(title = "Number of Accidents by Month", x = NULL, y = NULL, fill = "Number of Accidents")
```


Figure 8 Code:

```
# bin accident times and make factors
accidents <- accidents %>%
  mutate(Time = case_when(Hour >= 0 & Hour < 3 ~ "12AM - 3AM",
                           Hour >= 3 & Hour < 6 ~ "3AM - 6AM",
                           Hour >= 6 & Hour < 9 ~ "6AM - 9AM",
                           Hour >= 9 & Hour < 12 ~ "9AM - 12PM",
                           Hour >= 12 & Hour < 15 ~ "12PM - 3PM",
                           Hour >= 15 & Hour < 18 ~ "3PM - 6PM",
                           Hour >= 18 & Hour < 21 ~ "6PM - 9PM",
                           Hour >= 21 ~ "9PM - 12AM"),
         Time = factor(Time, levels = c("12AM - 3AM",
                                         "3AM - 6AM",
                                         "6AM - 9AM",
                                         "9AM - 12PM",
                                         "12PM - 3PM",
                                         "3PM - 6PM",
                                         "6PM - 9PM",
                                         "9PM - 12AM")),
         Severity = factor(Severity))

# make mosaic plot
ggplot(data=accidents) +
  geom_mosaic(aes(x=product(Severity, Time), fill=Severity)) +
  labs(title="Car Accident Severity by Time of Day",
       y="Severity",
       fill="Severity") +
  # color palette
  scale_fill_manual(values = c("1"="blue",
                              "2"="steelblue",
                              "3"="orange",
                              "4"="red")) +
  # no grey background
  theme_minimal() +
  # angle x-axis labels
  theme(axis.text.x=element_text(angle=45, hjust=1),
        plot.title=element_text(size=17, hjust=0.5)) # put title in middle
```

Figure 9 Code:

```
# extract time variables
accident <- accident %>%
  mutate(
    Start_Hour = hour(Start_Time),
    Start_Month = month(Start_Time, label = TRUE),
    Start_Day = wday(Start_Time, label = TRUE),
    Start_Year = year(Start_Time)
  )

# Table: Severity per Year
severity_time_table <- accident %>%
  filter(!is.na(Severity), !is.na(Start_Year)) %>%
  select(Start_Year, Severity) %>%
  table()

mosaicplot(severity_time_table,
  main = "Accident Severity by Year",
  xlab = "Year",
  ylab = "Severity Level",
  color = c("springgreen4", "gold2", "orange2", "tomato3"),
  cex.axis = 0.9,
  las = 1,
  border = "black")
old_par <- par(mar = c(5, 4, 4, 6)) # Increase right margin (last value)
colnames(severity_time_table) <- rep("", ncol(severity_time_table))
# Create mosaic plot
mosaicplot(severity_time_table,
  main = "Accident Severity by Year",
  xlab = "", # Remove x-axis label
  ylab = "", # Remove y-axis label
  color = c("springgreen4", "gold2", "orange2", "tomato3"),
  cex.axis = 0.9,
  las = 1,
  border = "black")

# Add legend outside the plot area
legend("right",
  legend = c("1", "2", "3", "4"),
  fill = c("springgreen4", "gold2", "orange2", "tomato3"),
  title = "Severity Level",
  cex = 0.8,
  bty = "n",
  inset = c(-0.14, 0),
  xpd = TRUE)|
```

Figure 10 Code:

```
# accident distribution waffle chart
ggplot(grouped_weather,
  aes(fill = Weather, values = Percentage)) +
geom_waffle(color = "white", size = 0.5, n_rows = 10) +
scale_fill_manual(values = c(
  Clear = "#EFB743",
  Cloudy = "#748b97",
  Rainy = "#4269D0",
  "Snow/Ice" = "#97BBF5",
  "Fog/Haze" = "#9498D8",
  Thunderstorm = "#984EA3",
  Other = "#BDBDBD"
)) +
labs(
  title = "Distribution of Traffic Accidents by Weather Condition",
  fill = "Weather Condition"
) +
theme_void() +
theme(
  plot.title = element_text(size = 18, face = "bold"),
  legend.title = element_text(size = 16),
  legend.text = element_text(size = 14)
)
```

Figure 11 Code:

```
# severity normalized stacked bar chart
ggplot(severity_weather,
  aes(x = Weather, y = Count,
    fill = factor(Severity))) +
geom_bar(stat = "identity",
  position = "fill") +
scale_y_continuous(labels = scales::percent) +
scale_fill_manual(values = c(
  "1" = "green",
  "2" = "gold",
  "3" = "orange",
  "4" = "red"
)) +
labs(
  title = "Severity Distribution Across Weather Conditions",
  x = "Weather Condition",
  y = "Percentage of Accidents",
  fill = "Severity"
) +
theme_minimal() +
theme(
  plot.title = element_text(size = 18, face = "bold")
)
```

Appendix D – Citations:

1. <https://www.cdc.gov/transportation-safety/about/index.html>
2. <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents?resource=download>